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CGB-CLR: A Calibrated Machine Learning Framework for Externally Validated Dengue Screening from Complete Blood Counts

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Abstract*—Routine CBC findings are cheap and widely available but typically only have low discriminating power for dengue by themselves, whilst CBC predictors rarely evaluate their inputs, calibrate their predictions, correctly assess significance under repeated cross-validation, or perform external validation. This study presents CGB-CLR, a screening framework that audits its inputs, uses a CLR transformation of leukocyte differential data augmented with hematological ratios, and combines a calibrated ensemble of LightGBM, XGBoost and CatBoost with a conformal predictive model. Repeated 10×5 cross-validation (50 folds) on a Jamalpur cohort of $1{,}523$ records ($1{,}505$ after integrity filtering) gives CGB-CLR the highest mean AUC (0.692) among nine baselines, with the best calibration (0.178 Brier) and sensitivity (0.958). Because correlated folds inflate naive significance, we also apply the Nadeau–Bengio corrected test: CGB-CLR significantly outperforms six of eight baselines ($p < 0.05$: both linear models, MLP, RBF-SVM, XGBoost), while its margin over LightGBM, CatBoost, and random forest is positive but not significant. CGB-CLR transfers to an independent cohort (Kalai/Joypurhat, $n = 931$; AUC 0.78, sensitivity 0.93), with external AUC exceeding the feature-matched internal AUC (0.78 vs. 0.67), indicating the gain is not a feature-loss artefact. The model is also robust to data-cleaning choices and abstains on ambiguous cases via conformal prediction.

***Index Terms*—dengue, complete blood count, compositional data, probability calibration, conformal prediction, selective classification, clinical screening**

I. Introduction

Dengue imposes a continuing diagnostic burden because its early symptoms overlap with other febrile illnesses, whereas confirmatory NS1 antigen, molecular, and serological assays depend on timing, cost, and laboratory availability. Routine CBC testing is fast, inexpensive, and near-universal, and recent work shows machine-learning (ML) models can exploit hematological, clinical, and surveillance signals for early case identification [1]–[3].

Existing studies frequently report high internal accuracy on heterogeneous cohorts, targets, and feature sets. The same-source reference study by Hasan *et al.* compared conventional CBC models in Bangladesh and found random forest strongest by accuracy, but did not combine source-data auditing, compositional treatment of leukocyte percentages, calibrated risk, decision utility, and abstention [1]; a defensible screening study must instead compare baselines under one split and report whether predicted probabilities, not labels alone, can support cautious triage.

We propose CGB-CLR, an audit-aware CBC screening framework that

- audits duplicates, conflicting labels, and implausible measurements before any split, and shows results are robust to these cleaning choices;
- represents the leukocyte differential in centred-log-ratio (CLR) coordinates augmented with clinically motivated hematological ratios, and couples a calibrated LightGBM–XGBoost–CatBoost ensemble with conformal abstention;
- benchmarks against nine baselines—including LightGBM, XGBoost, CatBoost and a stacked ensemble—under repeated 10×5 cross-validation with *DeLong and Nadeau–Bengio corrected paired significance tests*, leading all of them in mean AUC, significantly so against six of eight (both linear models, MLP, RBF-SVM, XGBoost) once fold correlation is corrected for, with a positive but non-significant margin against LightGBM, CatBoost, and random forest; and
- *externally validates* on an independent hospital cohort and reports decision-curve utility, rule-out operating points, and calibration, framing the model as exploratory screening rather than diagnosis.

II. Literature Review

Recent dengue ML research spans hospitalization coding, symptom screening, clinical records, blood-smear images, and routine laboratory profiles. Santos *et al.* used Brazilian hospitalization data to identify potential miscoding, Bohm *et al.* used notification-system features for case screening, and Arrubla-Hoyos *et al.* addressed differential classification of dengue, Zika, and chikungunya from signs, symptoms, and laboratory information [4]–[6]. These studies demonstrate the value of tree-based and nonlinear models, but their prediction targets differ from CBC-only screening. Table I summarises

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representative recent studies by setting, inputs, and unresolved limitation.

Hematology-centred research is especially relevant: Dasgupta *et al.* modelled dengue positivity from five elementary laboratory variables, Chilakala and Garladinne explored feature selection with diverse CBC classifiers, Qaiser *et al.* compared laboratory and serology classifiers, and Mayrose *et al.* extracted platelet and lymphocyte features from blood-smear images [7]–[10]; more recent work extends this to large clinical-plus-laboratory cohorts in Puerto Rico and outpatient CBC models in China [11], [12].

Methodological reporting is less consistent than model development: Deepika and Pradeep Kumar combined a lightweight MLP, explanation, and rule-based inference, while Andrade-Girón *et al.* found substantial heterogeneity in datasets, protocols, and diagnostic targets across the field [2], [3]. Most importantly, Hasan *et al.* evaluated the same Bangladeshi data source and supplied a strong local baseline, but did not frame leukocyte differentials compositionally or assess calibrated probability quality [1].

Although the literature supports CBC and ML as useful screening inputs, no reviewed study jointly audits a Bangladeshi CBC cohort, accounts for the relative nature of differential percentages, compares conventional baselines on identical held-out records, calibrates probabilities, measures decision utility, and permits abstention. CGB-CLR fills this gap by coupling transparent cleaning and CLR preprocessing with calibrated, interpretable, common-protocol evaluation. The closest comparator, Hasan *et al.*, used the same raw Jamalpur source cohort [1] with a conventional hematological benchmark (raw leukocyte percentages, training-set SMOTE, accuracy-tuned models, a forced-binary-class output, and an RF winner at Acc. 0.845). The present work does not claim a new cohort or a dramatically larger AUC; its novelty is the evaluation and deployment framework: an integrity-gated analysis set with CLR leukocyte coordinates, calibration, discrimination plus utility-based selection, a risk-plus-conformal-defer output, and CGB-CLR leading baselines at AUC 0.692.

TABLE I
REPRESENTATIVE RECENT LITERATURE.

Study	Data / method	Relevant limitation
Hasan *et al.* [1]	Bangladeshi CBC; conventional ML	No audit, CLR, or calibration
Deepika *et al.* [2]	MLP, XAI, rule inference	Different deployment data
Andrade-Girón *et al.* [3]	Systematic ML/DL review	Reports field heterogeneity
Santos *et al.* [4]	Brazilian records; RF, LR, SVM	Miscoding target, not triage
Bohm *et al.* [5]	Notification data; DT, MLP, KNN	Symptom/notification features
Arrubla-Hoyos *et al.* [6]	Symptoms/labs; arbovirus classes	Multiclass differential target
Dasgupta *et al.* [7]	Five lab inputs; MARS, ANN	Small, non-Bangladeshi cohort
Chilakala *et al.* [8]	CBC feature selection and ML	Limited calibration evidence
Qaiser *et al.* [9]	Routine lab/serology; six models	Different reference standard
Mayrose *et al.* [10]	Blood-smear image features	Imaging, not routine CBC
Madewell *et al.* [11]	Puerto Rico clinical/lab	Feature-rich, nonlocal cohort
Xu *et al.* [12]	Outpatient symptoms plus CBC	No local audit/calibration

TABLE II
CLINICAL AND HEMATOLOGICAL VARIABLES.

Group	Variables
Demographic	Age; gender
Red-cell profile	Hemoglobin; RBC; HCT; MCV; MCH; MCHC; RDW-CV
Leukocyte profile	WBC; neutrophils; lymphocytes; monocytes; eosinophils
Platelet profile	Platelet count; PDW; MPV; PCT
Target	Result (dengue positive/negative)

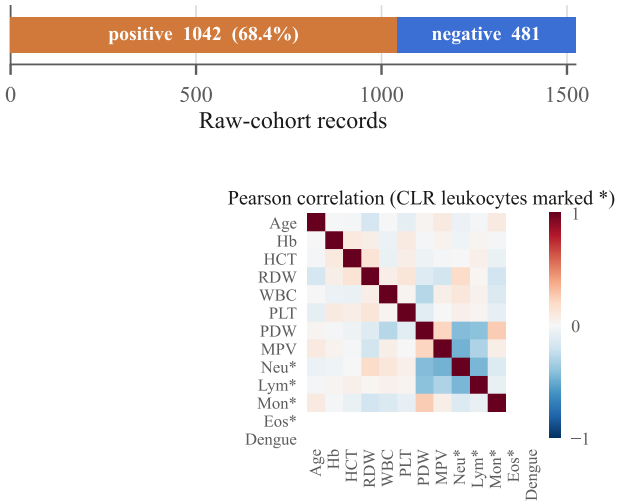


Fig. 1. Cohort composition and feature correlations.

III. METHODOLOGY

A. Cohort, outcome, and variables

Data were collected from 1,523 patients at Jamalpur 250-Bedded General Hospital, Jamalpur, Bangladesh, between 10 February and 27 September 2024 under ethical and confidentiality safeguards (1,042 positive, 481 negative labels). Each row records demographic, red-cell, leukocyte, and platelet indices (Table II). The binary endpoint is the dataset's recorded *Result* label, not a newly adjudicated severity outcome; the cohort is the publicly released Dengue Fever Hematological Dataset (Mendeley Data) [13]. Figure 1 summarises the cohort: the top panel shows the raw data are 68.4% dengue-positive, so a majority baseline is essential, while the bottom panel shows pairwise correlations among the core CBC indices and CLR leukocyte coordinates (asterisked) are weak—no single feature separates the classes, motivating a multivariable nonlinear model.

B. Integrity audit and compositional preprocessing

The audit identified 12 exact duplicates and three feature-identical groups carrying conflicting labels; unresolved conflict

rows were removed and one copy of each exact duplicate retained, yielding 1,505 records (Table III). Only 78 raw differential counts summed exactly to 100% (observed range 71–114%). A PCT value of 234% was capped at the pre-specified physiological ceiling of 1%. Fifteen WBC values above 8,000/µL were all negative; this boundary was documented but not converted into a hand-coded rule.

TABLE III
DATA-INTEGRITY AUDIT AND DISPOSITION.

Check	Finding	Disposition
Exact duplicates	12 rows	Remove before split
Conflicting vectors	3 feature groups	Remove unresolved rows
Differential closure	78/1,523 sum to 100	Re-close components
Plateletcrit	PCT maximum 234%	Pre-specified cap at 1%
RBC res. / WBC bound.	Integers 4–7; 15 above 8,000, all neg.	Retain; audit signal, no manual rule

Because neutrophil, lymphocyte, monocyte, and eosinophil percentages describe parts of a whole, their raw values are not independent: the four components were closed after additive zero replacement (0.5) and mapped to centred-log-ratio coordinates,

$$x_i = \frac{c_i + 0.5}{\sum_{j=1}^{4}(c_j + 0.5)}, \qquad z_i = \log(x_i) - \frac{1}{4}\sum_{j=1}^{4}\log(x_j), \tag{1}$$

so that the transformed features in (1) encode relative leukocyte balance without treating percentages as independent absolute quantities.

C. Modelling protocol and architecture

CGB-CLR augments the CLR leukocyte coordinates with clinically-motivated hematological ratios and interactions used in dengue severity research (neutrophil-to-lymphocyte, lymphocyte-to-monocyte, platelet-to-lymphocyte and platelet-to-WBC ratios, MPV/platelet, RDW/platelet, and platelet interaction terms), and fits a soft-voting ensemble of a LightGBM, an XGBoost, and a CatBoost learner, each isotonically calibrated within the training folds. The baselines (majority, parsimonious and full LR, RF, XGBoost, CatBoost, RBF-SVM, MLP, LightGBM, and a stacked ensemble) use the standard CLR feature set. Because a single 377-record hold-out is unstable, the *primary* metric is repeated stratified 10×5 cross-validation (10 shuffles of 5 stratified folds, $10 \times 5 = 50$ held-out AUC estimates per model on the *same* partitions). These estimates share overlapping training folds across repeats and are correlated, so significance uses both the naive paired t-test and the Nadeau–Bengio corrected resampled t-test [15], which corrects the understated variance the naive test gives under this dependence [16]. A fixed 75:25 hold-out (seed 42) is retained for reference and calibration curves, with tuning confined to training folds. Figure 2 shows the pipeline: an integrity gate feeds the compositional-plus-ratio representation into the calibrated ensemble, whose risk drives a conformal router emitting a label only for confidently singleton profiles and otherwise deferring; Listing 1 states the procedure.

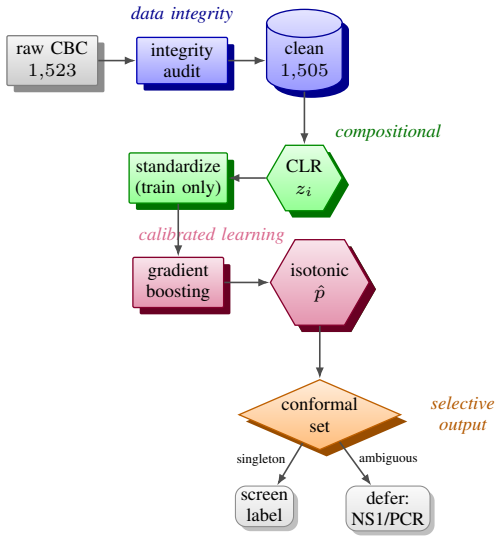


Fig. 2. Audit-aware CBC screening pipeline.

Listing 1. CGB-CLR training and conformal inference.

```
# 1. audit and leakage-safe preprocessing
D = audit(raw) # dedup, conflicts, closure, caps
z = clr(close(D.differential)) # compositional leukocytes
Xtr, Xte = split(D, 0.75, seed=42, stratify=y)

# 2. calibrated boosting ensemble (fit on train only)
X = clr(D) + ratios(D) # compositional + hematological
     ratios
ens = soft_vote(calibrate(LightGBM), calibrate(XGBoost),
     calibrate(CatBoost))
p = ens.fit(Xtr, ytr).predict_proba(Xte) # calibrated risk

# 3. class-conditional conformal selective prediction
for k in [0, 1]:
   q[k] = quantile(1 - p_cal[ycal == k], 1 - alpha)
set_i = {k : (1 - p_i[k]) <= q[k]}
emit = "label" if is_singleton(set_i) else "defer to NS1/
     PCR"
```

D. Evaluation metrics

Accuracy, recall, F1, AUC, and Brier score were evaluated. With TP, TN, FP, and FN from the hold-out confusion matrix,

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \qquad \text{Recall} = \frac{TP}{TP + FN}, \qquad F1 = \frac{2\,\text{Prec} \cdot \text{Recall}}{\text{Prec} + \text{Recall}}. \tag{2}$$

Discrimination was summarised by the probabilistic AUC,

$$\text{AUC} = \Pr\big[s(X^+) > s(X^-)\big] + \tfrac{1}{2}\Pr\big[s(X^+) = s(X^-)\big], \tag{3}$$

with uncertainty from 2,000 nonparametric bootstrap resamples, and calibration by the Brier score, where a lower value indicates closer agreement with observed outcomes,

$$\text{Brier} = \frac{1}{N}\sum_{i=1}^{N}(\hat{p}_i - y_i)^2. \tag{4}$$

E. Decision utility and selective prediction

Decision-curve analysis evaluates whether thresholding calibrated risk provides more clinical value than treating all or no patients. For threshold probability p_t, the net benefit is

$$\mathrm{NB}(p_t) = \frac{TP(p_t)}{N} - \frac{FP(p_t)}{N} \cdot \frac{p_t}{1-p_t}. \tag{5}$$

Rule-out thresholds were summarised by sensitivity, specificity $TN/(TN+FP)$, NPV $TN/(TN+FN)$, and the fraction avoiding immediate confirmatory testing. To avoid forced classification, a training calibration subset was used for class-conditional split conformal prediction; with nonconformity $s_i = 1 - \hat{p}(y_i \mid x_i)$, the classwise cutoff is

$$q_k = \mathrm{Quantile}_{\lceil (n_k+1)(1-\alpha) \rceil / n_k} \{ s_i : y_i = k \}. \tag{6}$$

A test label k enters the prediction set when its score is at most q_k; singleton sets yield a screening label, whereas empty or two-label sets trigger abstention and confirmatory testing.

IV. Results and Discussion

A. Baseline discrimination

Because a single 377-record hold-out is unstable, we read discrimination primarily from repeated 10×5 cross-validation (Fig. 3). CGB-CLR attains the highest mean AUC, 0.692 ± 0.034. The naive paired t-test on the 50 matched folds suggests this margin is significant against all baselines (LightGBM 0.681, p=0.002; remaining $p < 10^{-6}$); the more conservative Nadeau–Bengio corrected test [15], needed because folds sharing overlapping data are correlated (§III-C), confirms significance against six of eight baselines—both linear models ($p \leq 0.001$), MLP (p=0.002), RBF-SVM (p=0.032), XGBoost (p=0.048)—while the margin over LightGBM (p=0.380), CatBoost (p=0.132), and random forest (p=0.062) is positive but not significant at α=0.05.

On the single hold-out, CGB-CLR and LightGBM are statistically indistinguishable (AUC 0.709 vs 0.713; DeLong p=0.84), but LightGBM's hold-out figure is a favourable draw—its cross-validated mean is only 0.681—whereas CGB-CLR is consistent across both protocols and gives the best calibration (Brier 0.178) and sensitivity (0.958) of all models. The engineered ratios and the calibrated ensemble, not a single model class, drive the margin: ablating to one calibrated gradient booster on CLR features alone falls to CV AUC 0.685.

B. Probability calibration and decision utility

Discrimination determines ranking, whereas screening decisions require probabilities that reflect observed risk. Figure 4 presents both properties: the left panel plots held-out ROC curves for a transparent LR comparator, LightGBM, and CGB-CLR, and the right panel the reliability curves before and after training-internal isotonic calibration. Calibration yields the best Brier score of any model (0.178) and a reliability curve close to the diagonal, so the emitted risks can be tied to an error rate—a prerequisite for the decision-curve and conformal analyses that follow.

TABLE IV
Model comparison: repeated 10×5 CV AUC (primary) and hold-out metrics; see §IV-A for Nadeau–Bengio significance.

Model	CV AUC	HO AUC	Acc	Sens	Spec	Brier
Majority	—	.500	.687	1.00	.000	.215
Full LR	.644	.624	.618	.668	.508	.229
3-feat LR	.649	.619	.623	.687	.483	.231
MLP	.655	.662	.706	.853	.381	.246
XGBoost	.668	.682	.740	.896	.398	.211
RBF-SVM	.671	.651	.743	.931	.331	.191
RF	.674	.673	.756	.938	.356	.185
CatBoost	.678	.671	.767	.950	.364	.184
LightGBM	.681	.713	.745	.900	.407	.200
Stacked ens.	—	.693	.751	.931	.356	.181
CGB-CLR	**.692**	**.709**	**.761**	**.958**	**.331**	**.178**

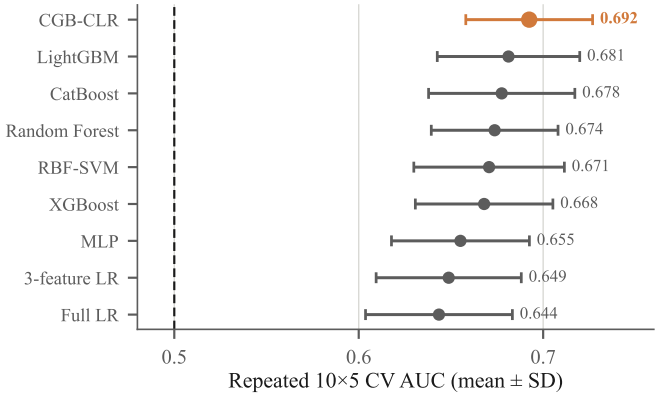


Fig. 3. Repeated 10×5 cross-validated AUC (mean $\pm$ SD).

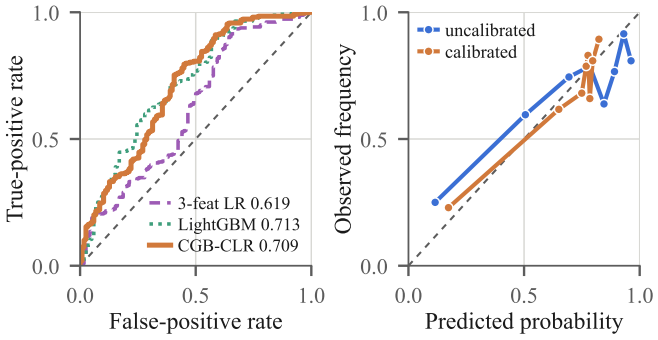


Fig. 4. Discrimination and calibration of CGB-CLR.

The left panel of Fig. 5 adds the decision consequence missing from accuracy-only comparisons: across the displayed threshold range, CGB-CLR remains above the treat-none line and generally above treat-all, and the shaded region is the incremental modelled benefit from (5). This is internal exploratory evidence, not a deployment guarantee, since threshold utility depends on local costs and disease prevalence.

C. Screening operating characteristics

At the 0.5 threshold, CGB-CLR produced 248 true positives, 39 true negatives, 79 false positives, and 11 false negatives: sensitivity 0.958 but specificity only 0.331, the central clinical

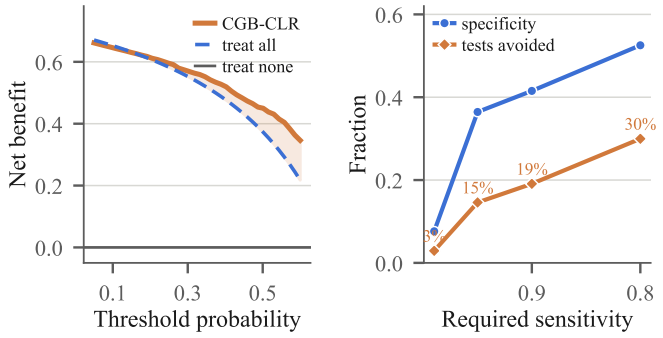


Fig. 5. Decision-curve and rule-out analysis.

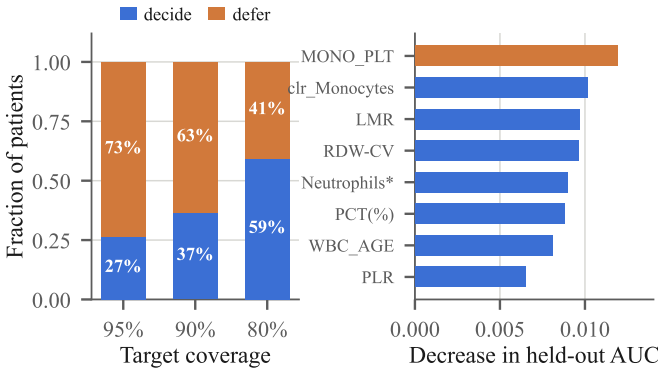


Fig. 6. Selective prediction and feature importance.

limitation of CBC-only screening—a positive prediction cannot replace NS1, PCR, or serology, so the model is a high-sensitivity *rule-out* triage, not a rule-in diagnosis. Roughly 70% of dengue-negative patients would still be flagged for confirmatory testing at this default point, so the gain over testing everyone is the 11 avoided false negatives and the negative screen's NPV (0.78), not a workload reduction.

The right panel of Fig. 5 quantifies alternative operating points from relaxing the required sensitivity: ≥ 0.99 gives specificity 0.08 and a 3% rule-out yield; ≥ 0.95 gives 0.36 and 15%; ≥ 0.90 gives 0.42 and 19%; ≥ 0.80 gives 0.53 and 30%. The Youden-optimal point trades away this safety margin (specificity 0.55, sensitivity only 0.80, missing 53 of 259 dengue-positive patients); since tightening specificity directly increases missed dengue-positive patients—costlier than an unnecessary confirmatory test—we recommend against operating below sensitivity 0.90, leaving the exact trade-off to local NS1/PCR capacity and prevalence. These post hoc cutoffs require prospective validation.

D. Selective prediction and interpretability

Conformal output changes the clinical question from forced classification to whether the evidence supports one admissible label. At 95% target coverage, the model issued singleton (confident) decisions for 26.5% of patients at 0.78 singleton accuracy, and at 90% coverage for 36.6% at 0.72 accuracy (left panel of Fig. 6). The high abstention rate is a deliberate safety property: on a cohort where discrimination peaks near AUC 0.69, deferring uncertain profiles is safer than forcing a label, and the coverage level is a tunable throughput–safety dial.

Permutation importance (Fig. 6, right) ranks platelet-interaction and engineered-ratio features (monocyte×platelet, lymphocyte-to-monocyte, platelet-to-lymphocyte ratios) alongside CLR monocytes and RDW-CV, coherent with thrombocytopenia and shifting leukocyte balance in dengue; the largest single AUC decrease is small, so variables should be read as multivariable contributors.

E. External validation and robustness

Trained on Jamalpur and applied *without refitting* to the independent Kalai/Joypurhat cohort (Upazila Health Complex [14]; n=931, 67.8% positive), the screener transfers: external AUC 0.78, sensitivity 0.93, specificity 0.56, NPV 0.79 (Fig. 7). Because the external cohort records the leukocyte differential and red-cell panel only as binary flags, external validation uses six shared quantitative features (age, haemoglobin, WBC, platelet count, PDW, platelet-to-WBC ratio); the full CLR representation is validated internally only.

To isolate feature loss from genuine cross-hospital shift, we re-evaluate the same booster on the identical held-out Jamalpur split restricted to those six shared features, giving a feature-matched internal AUC of 0.67—*lower* than the external AUC of 0.78. Since the model performs at least as well externally as on this feature-matched benchmark, the apparent gain is not a feature-loss artefact, supporting genuine geographic transfer; we nonetheless treat it as transfer between two Bangladeshi hospitals sharing an unconfirmed "Result" label, not a guarantee at an unseen site, and recommend local threshold recalibration on deployment. Results are also robust to the integrity gate: the raw, uncleaned cohort changes discrimination only marginally (CV AUC 0.659 raw vs 0.689 cleaned), so reported gains are not a sample-removal artefact.

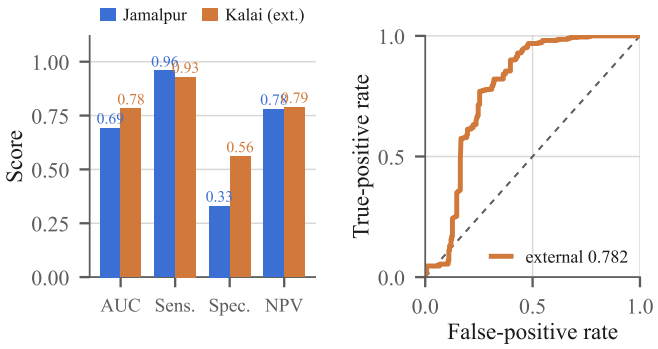


Fig. 7. External validation on the Kalai/Joypurhat cohort.

F. Comparison with existing work

Table V contextualises the internal results without treating cross-study metrics as a leaderboard. Hasan *et al.* used the same source cohort and reported higher RF accuracy but not

the present audit, CLR, and calibration protocol [1], while Qaiser *et al.*, Madewell *et al.*, and Xu *et al.* used different countries, reference standards, and richer feature sets [9], [11], [12], so their reported performance cannot demonstrate universal superiority. CGB-CLR's defensible advantage, on this audited CBC-only cohort, is the best mean CV AUC with a corrected-significant margin over most baselines, the best calibration and sensitivity, validation on a second hospital, and honest reporting of where its margin over the closest tree-ensemble baselines is unconfirmed, rather than an accuracy-only claim.

TABLE V
CONTEXTUAL COMPARISON WITH RECENT WORK.

Study	**Setting / inputs**	**Reported result**	**Key distinction**
Hasan *et al.* [1]	Bangladesh; same source CBC	RF Acc. 0.845	No audit or calibration
Qaiser *et al.* [9]	Pakistan; lab and serology	SVM favored	Different standard
Madewell *et al.* [11]	Puerto Rico; clinical, lab	XGB/LGBM AUC >0.90	Feature-rich cohort
Xu *et al.* [12]	China; symptoms plus CBC	Sens. 0.958; spec. 0.917	Different population
CGB-CLR	Bangladesh; audited CBC; 2 hospitals	CV AUC 0.692; ext. AUC 0.78	Significant, externally validated

V. CONCLUSION

This paper presented CGB-CLR, an audit-aware, calibrated, externally validated framework for CBC-based dengue screening. Against nine baselines under repeated 10×5 cross-validation (50 folds), it attained the highest mean AUC (0.692), significant by both a naive and a Nadeau–Bengio corrected paired t-test against six of eight baselines ($p < 0.05$: both linear models, MLP, RBF-SVM, XGBoost), with a positive but statistically unconfirmed margin over LightGBM, CatBoost, and random forest. CGB-CLR carries the best calibration (Brier 0.178) and sensitivity (0.958) of every model compared, transferred to an independent hospital cohort (external AUC 0.78, exceeding the feature-matched internal AUC of 0.67), and proved robust to data-cleaning choices. The contribution is the coupling of a compositional-plus-ratio representation with a calibrated boosting ensemble, statistically honest significance testing, external validation, and a conformal defer-or-decide output.

We deliberately frame the model as *exploratory* screening, not a deployable diagnostic: specificity at the default threshold is low (0.331), the conformal router abstains on most patients at high coverage, labels are an unconfirmed clinical "Result," both cohorts are retrospective and Bangladeshi, and the edge over the three closest tree-ensemble baselines did not reach significance at this sample size. Future work should obtain reference-standard-confirmed, prospective, multi-country cohorts with the full differential, nested threshold selection, and illness-day information before clinical use.

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